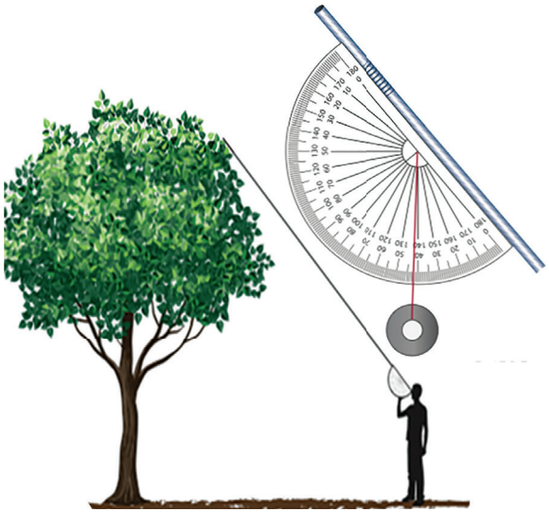


string should hang down freely, crossing the protractor portion of the clinometers. Read the angle shown, and subtract from 90° to find your angle of vision from your eye to the top of the tree (it can be helpful here to have an assistant to read the measurement while we look through the straw). We record our results on a paper. From the spot, the clinometers (read by assistant) show 55°. Subtracting from 90° that indicated that we looked at an angle of 35° to the top of the tree.

Step 3: Measure Distance

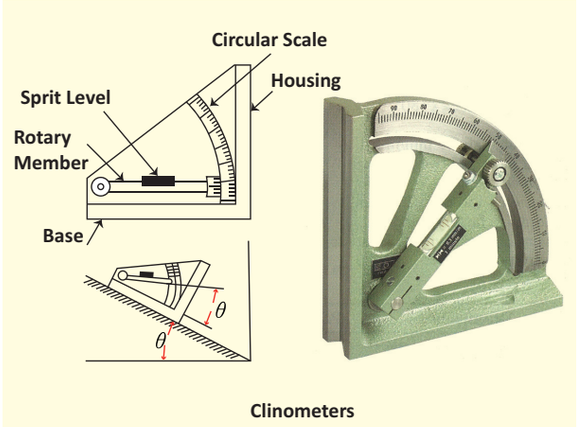
Once we have our angle of vision, we can use tape measure to find the distance from the spot we are standing to the base of the object we are measuring (an assistant comes in handy here, too). We must know how far away we are to accurately calculate the height. Our spot was 15.6 meters from the base of the tree we measured.

Step 4: Find Your Eye-height

The last piece of data you need to calculate the height of the object is the height from the ground to our eye (our eye-height). We must have the assistant help us measure this using our tape measure. Our eye height was recorded for this example as 1.64 metres.

Step 5: Draw a Picture

In calculating the height of the object you just measured, we find it helpful to begin by

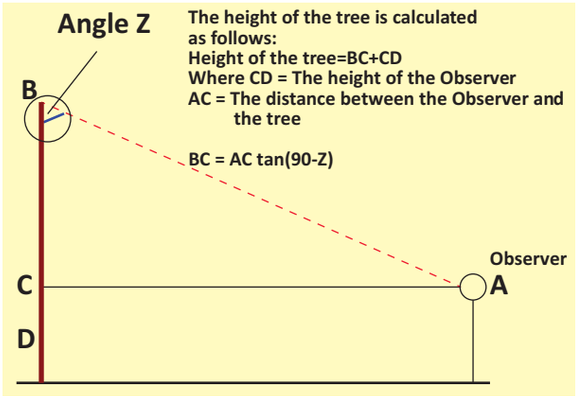


Clinometers

drawing a picture and labeling it with all of the information we have.

Step 6: Model as a Triangle

The next step is to simplify our drawing to model our system as a right triangle. Label our triangle with the angle we read on our clinometers as well as the distance you were standing from the object (we don't need the eye-height just yet).



We can find x in this triangle (which represents the portion of the height from eye-level up) by using some basic trigonometry, specifically the tangent ratio of the triangle:

tan(**angle**) = x / **distance**

Multiply by the distance on both sides and you get:

x = tan(**angle**) * **distance**

Use a calculator to multiply these together and get a decimal value.

In the example:

tan(35°) = x / 15.6

x = tan(35°) * 15.6

x = 10.92 metres

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